Marmara University
Electrical and Electronics Engineering
Probability and Statistics
Problem Sets with Solutions #7

[1] If

$$P_{X,Y}[i,j] = \begin{cases} 1/6, & i=0, j=0 \\ 1/3, & i=0, j=1 \\ 1/3, & i=1, j=0 \\ 1/6, & i=1, j=1 \end{cases}$$

find $P_{Y|X}$ and $P_{X|Y}$

Solution:

$P_X[0] = \frac{1}{6}+\frac{1}{3} = \frac{1}{2}$ $\quad P_Y[0] = \frac{1}{6}+\frac{1}{3} = \frac{1}{2}$

$P_X[1] = \frac{1}{3}+\frac{1}{6} = \frac{1}{2}$ $\quad P_Y[1] = \frac{1}{3}+\frac{1}{6} = \frac{1}{2}$

$$P_{Y|X}[j|i] = \frac{P_{X,Y}[i,j]}{P_X[i]}$$

$$P_{Y|X}[0|0] = \frac{P_{X,Y}[0,0]}{P_X[0]} = \frac{1/6}{1/2} = \frac{1}{3}$$

$$P_{Y|X}[0|1] = \frac{P_{X,Y}[1,0]}{P_X[1]} = \frac{1/3}{1/2} = \frac{2}{3}$$

$$P_{Y|X}[1|0] = \frac{P_{X,Y}[0,1]}{P_X[0]} = \frac{1/3}{1/2} = \frac{2}{3}$$

$$P_{Y|X}[1|1] = \frac{P_{X,Y}[1,1]}{P_X[1]} = \frac{1/6}{1/2} = \frac{1}{3}$$

Similarly

$$P_{X|Y}[0|0] = \frac{1/6}{1/2} = \frac{1}{3}$$

$$P_{X|Y}[0|1] = \frac{1/3}{1/2} = \frac{2}{3}$$

$$P_{X|Y}[1|0] = \frac{1/3}{1/2} = \frac{2}{3}$$

$$P_{X|Y}[1|1] = \frac{1/6}{1/2} = \frac{1}{3}$$

□

[2] A joint PMF $P_{X,Y}[i,j]$ has the values shown in the table below. Determine the conditional PMF $P_{Y|X}$. Are the random variables independent?

	$j=1$	$j=2$	$j=3$
$i=1$	1/10	1/10	2/10
$i=2$	1/20	1/20	1/10
$i=3$	3/10	1/20	1/20

Solution:

$$P_{Y|X}[j|i] = \frac{P_{X,Y}[i,j]}{P_X[i]}, \quad P_X[1] = \frac{1}{10}+\frac{1}{10}+\frac{2}{10} = \frac{4}{10}$$

$$P_X[2] = \frac{1}{20}+\frac{1}{20}+\frac{1}{10} = \frac{4}{20} = \frac{2}{10}$$

$$P_X[3] = \frac{3}{10}+\frac{1}{20}+\frac{1}{20} = \frac{8}{20} = \frac{4}{10}$$

$$P_{Y|X}[j|1] = \frac{\frac{1}{10}}{\frac{4}{10}}, \frac{\frac{1}{10}}{\frac{4}{10}}, \frac{\frac{2}{10}}{\frac{4}{10}} = \frac{1}{4}, \frac{1}{4}, \frac{1}{2} \quad (j=1,2,3)$$

$$P_{Y|X}[j|2] = \frac{\frac{1}{20}}{\frac{2}{10}}, \frac{\frac{1}{20}}{\frac{2}{10}}, \frac{\frac{1}{10}}{\frac{2}{10}} = \frac{1}{4}, \frac{1}{4}, \frac{1}{2} \quad (j=1,2,3)$$

$$P_{Y|X}[j|3] = \frac{\frac{3}{10}}{\frac{4}{10}}, \frac{\frac{1}{20}}{\frac{4}{10}}, \frac{\frac{1}{20}}{\frac{4}{10}} = \frac{3}{4}, \frac{1}{8}, \frac{1}{8} \quad (j=1,2,3)$$

$P_Y[1] = \frac{1}{10}+\frac{1}{20}+\frac{3}{10} = \frac{9}{20}$ $\quad P_Y[3] = \frac{2}{10}+\frac{1}{10}+\frac{1}{20} = \frac{7}{20}$

$P_Y[2] = \frac{1}{10}+\frac{1}{20}+\frac{1}{20} = \frac{4}{20}$

$P_{Y|X} \neq P_Y$, X and Y are not independent!

[3] If X and Y have the joint PMF $P_{X,Y}[i,j] = \left(\frac{1}{2}\right)^{i+1} e^{-\lambda} \cdot \frac{\lambda^j}{j!}$ $i = 0,1,\ldots$; $j = 0,1,\ldots$ find $E_{Y|X}[Y|i]$ for all i.

Solution: $P_{Y|X}[j|i] = \dfrac{P_{X,Y}[i,j]}{P_X[i]}$

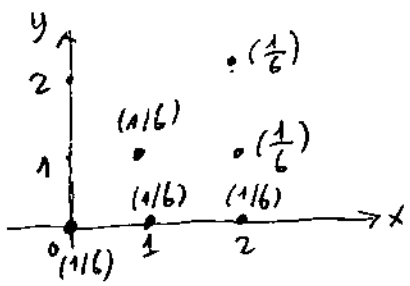
$$P_X[i] = \sum_{j=0}^{\infty} P_{X,Y}[i,j] = \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^{i+1} e^{-\lambda} \frac{\lambda^j}{j!} = \left(\frac{1}{2}\right)^{i+1} \cdot e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \left(\frac{1}{2}\right)^{i+1} \cdot e^{-\lambda} \cdot e^{\lambda} = \left(\frac{1}{2}\right)^{i+1}$$

$$P_{Y|X}[j|i] = \frac{\left(\frac{1}{2}\right)^{i+1} \cdot e^{-\lambda} \cdot \frac{\lambda^j}{j!}}{\left(\frac{1}{2}\right)^{i+1}} = e^{-\lambda} \cdot \frac{\lambda^j}{j!}$$ ($P_{Y|X}$ is a Poisson random variable!)

Hence $E_{Y|X}[Y|i] = \lambda$ (Expected value of Poisson r.v is λ) ☒

[4] Find $E[Y|X]$ if the joint PMF $P_{X,Y}[i,j]$ is uniformly distributed over the points $S_{X,Y} = \{(0,0), (1,0), (1,1), (2,0), (2,1), (2,2)\}$.

Solution:



$P_X[0] = \frac{1}{6}$

$P_X[1] = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$

$P_X[2] = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6}$

$P_{Y|X}[0|0] = 1$ $\quad P_{Y|X}[0|1] = 1/2$ $\quad P_{Y|X}[0|2] = 1/3$

$P_{Y|X}[1|0] = 0$ $\quad P_{Y|X}[1|1] = 1/2$ $\quad P_{Y|X}[1|2] = 1/3$

$P_{Y|X}[2|0] = 0$ $\quad P_{Y|X}[2|1] = 0$ $\quad P_{Y|X}[2|2] = 1/3$

$$E_{Y|X}[Y|0] = \sum_{j=0}^{2} j \cdot P_{Y|X}[j|0] = 0.1 + 1.0 + 2.0 = 0$$

$$E_{Y|X}[Y|1] = \sum_{j=0}^{2} j\, P_{Y|X}[j|1] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} + 2 \cdot 0 = \frac{1}{2}$$

$$E_{Y|X}[Y|2] = \sum_{j=0}^{2} j \cdot P_{Y|X}[j|2] = 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 2\,\frac{1}{3} = 1$$ ☒

[5] Prove that $\operatorname{var}(Y|X_i) = E_{Y|X}[Y^2|X_i] - (E_{Y|X}[Y|X_i])^2$

Solution: $\operatorname{var}(Y|X_i) = \sum_j (y_j - E_{Y|X}[Y|X_i])^2 \cdot P_{Y|X}[y_j|X_i]$

$$= \sum_j y_j^2 \cdot P_{Y|X}[y_j|X_i] - 2 \sum_j y_j E_{Y|X}[Y|X_i] \cdot P_{Y|X}[y_j|X_i] + \sum_j (E_{Y|X}[Y|X_i])^2 \cdot P_{Y|X}[y_j|X_i]$$

$$= E_{Y|X}[Y^2|X_i] - 2 E_{Y|X}[Y|X_i] \underbrace{\sum_j y_j P_{Y|X}[y_j|X_i]}_{E_{Y|X}[Y|X_i]} + (E_{Y|X}[Y|X_i])^2 \underbrace{\sum_j P_{Y|X}[y_j|X_i]}_{=1}$$

$$= E_{Y|X}[Y^2|X_i] - 2 (E_{Y|X}[Y|X_i])^2 + (E_{Y|X}[Y|X_i])^2$$

$$= E_{Y|X}[Y^2|X_i] - (E_{Y|X}[Y|X_i])^2$$ ☒

[6] In this problem we prove that $P_M[i] = e^{-\lambda}\frac{\lambda^i}{i!}$ and $P_{Y|M}[j|i] = \binom{i}{j} p^j (1-p)^{i-j}$

then $P_Y[j] = e^{-\lambda p}\cdot\frac{(\lambda p)^j}{j!}$

Solution: $\phi_Y(w) = E_Y[e^{jwY}] = E_M[\overbrace{E_{Y|M}[e^{jwY}|M]}^{\text{binomial r.v}}]$ ← characteristic function of binomial r.v.

$$= E_M[\underbrace{[pe^{jw}+(1-p)]}_{a}^{M}]$$

$$= \sum_{i=0}^{\infty} a^i \cdot P_M[i]$$

$$= \sum_{i=0}^{\infty} a^i \cdot e^{-\lambda}\cdot\frac{\lambda^i}{i!} = e^{-\lambda}\sum_{i=0}^{\infty}\frac{(a\lambda)^i}{i!} = e^{-\lambda}\cdot e^{a\lambda} \quad \lambda(a-1)$$

$$= e^{\lambda[pe^{jw}+(1-p)-1]}$$

$$= e^{\lambda p(e^{jw}-1)}$$

This is the characteristic function of Poisson r.v with parameter λp.

Hence, $P_Y[j] = e^{-\lambda p}\cdot\frac{(\lambda p)^j}{j!}$, $j = 0,1,2,\ldots$ ☒

[7] In this problem, we prove that the best predictor is given by the conditional mean $E_{Y|X}[Y|X_i]$. To prove this we let the predictor be $\hat{Y} = g(X)$ and minimize

$$E_{X,Y}[(Y-g(X))^2] = \sum_i\sum_j (y_j - g(x_i))^2 \cdot P_{X,Y}[x_i, y_j] = \sum_i\left[\sum_j (y_j-g(x_i))^2 \cdot P_{Y|X}[y_j|x_i]\right]\cdot P_X[x_i]$$

This is equivalent to minimizing

$$\sum_j (y_j - g(x_i))^2 P_{Y|X}[y_j|x_i]$$

Since $P_X[x_i]$ is nonnegative and we can choose a different value of $g(x_i)$ for each x_i. Prove that this is minimized for $g(x_i) = E_{Y|X}[Y|x_i]$.

Solution: Let $g(x_i) = c$.

$$h(c) = \sum_j (y_j - c)^2 P_{Y|X}[y_j|x_i]$$

$$\frac{dh(c)}{dc} = -2\sum_j (y_j - c)\cdot P_{Y|X}[y_j|x_i] = 0$$

$$\Rightarrow \sum_j y_j \cdot P_{Y|X}[y_j|x_i] = c\underbrace{\sum_j P_{Y|X}[y_j|x_i]}_{=1}$$

$$\Rightarrow c = \underbrace{\sum_j y_j P_{Y|X}[y_j|x_i]}_{E_{Y|X}[Y|x_i]}$$

$$\Rightarrow \boxed{c = E_{Y|X}[Y|x_i]}.$$

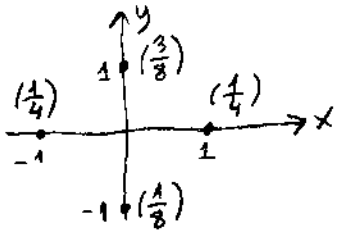
☒

[8] X and Y have the joint PMF

$$P_{X,Y}[i,j] = \begin{cases} 1/4, & (i,j) = (-1,0) \\ 1/8, & (i,j) = (0,-1) \\ 3/8, & (i,j) = (0,1) \\ 1/4, & (i,j) = (1,0) \end{cases}$$

We wish to predict Y based on our knowledge of the outcome of X. Find the optimal predictor. Also, find the optimal <u>linear</u> predictor and compare your results.

<u>Solution:</u>



$$\hat{Y} = E_{Y|X}[Y|X_i] = \sum_j y_j \, P_{Y|X}[y_j|X_i]$$

$$P_X[0] = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$P_X[1] = \frac{1}{4}$$

$$P_X[-1] = \frac{1}{4}$$

$P_{Y|X}[-1|-1] = 0$ $\quad P_{Y|X}[-1|0] = 1/4$ $\quad P_{Y|X}[-1|1] = 0$

$P_{Y|X}[0|-1] = 1$ $\quad P_{Y|X}[0|0] = 0$ $\quad P_{Y|X}[0|1] = 1$

$P_{Y|X}[1|-1] = 0$ $\quad P_{Y|X}[1|0] = 3/4$ $\quad P_{Y|X}[1|1] = 0$

if $X_i = -1 \Rightarrow \hat{Y} = E_{Y|X}[Y|-1] = \sum_{j=-1}^{1} j \cdot P_{Y|X}[j|-1] = (-1)(0) + 0.(1) + 1.(0) = 0$

if $X_i = 0 \Rightarrow \hat{Y} = E_{Y|X}[Y|0] = \sum_{j=-1}^{1} j \cdot P_{Y|X}[j|0] = (-1)\left(\frac{1}{4}\right) + 0.(0) + 1.\left(\frac{1}{4}\right) = \frac{1}{2}$

if $X_i = 1 \Rightarrow \hat{Y} = E_{Y|X}[Y|1] = \sum_{j=-1}^{1} j \cdot P_{Y|X}[j|1] = (-1).0 + 0.(1) + (1).(0) = 0.$

optimal linear predictor $\hat{Y} = E_Y[Y] + \frac{cov(X,Y)}{var(X)}(X - E_X[X])$

$E_X[X] = 0$, $E_Y[Y] = 1/4$, $var(X) = 1/2$, $cov(X,Y) = 0$.

Hence, the optimal linear predictor is $\hat{Y} = E_Y[Y] = 1/4$.

